
Database Systems (CSC-202)Credit Hours: **(Theory) 3**Pre-requisite: **Intro to ICT**Lectures: **Sec A** Mondays & Tuesdays (11:00AM to 12:30PM)**Sec B** Tuesdays & Thursdays (9:30AM to 11:00AM)**Sec C** Monday (2:00PM to 3:30PM) & Thursdays (11:00AM to 12:30PM)Instructor: **Dr. Natash Ali Mian**Semester: **Fall 2024**

Course Objective:

This is an introductory course in Database management. It links data and processes effectively and efficiently.

You will learn how databases are used to store, manipulate, and retrieve data in different types of organizations. It covers various technologies used to build and manage databases and how different applications are dependent upon databases today. The course explains how to build robust and flexible databases. Students will learn about the essential skills to design effective databases and to retrieve data through SQL queries.

Learning Outcomes:

1. To learn about the limitations of traditional file processing systems.
2. To be able to model an application's data requirements using conceptual modeling tools like ER diagrams and design database schemas based on the conceptual model.
3. To identify functional dependencies and to resolve database anomalies by normalizing database tables.
4. To use Structured Query Language (SQL) for database definition and manipulation in any DBMS
5. To introduce students with transaction management concepts
6. To introduce students with No SQL databases
7. To introduce students with latest topics and trends in databases

Text Books:

- A. Database Systems: A Practical Approach to Design, Implementation, and Management, 6 th Edition by Thomas Connolly and Carolyn Begg, Pearson
- B. Fundamentals of Database Systems 7th Edition by Ramez Elmasri, Shamkant B. Naathe
- C. Database System Concepts, 7th Edition by Avi Silberschatz, Henry F. Korth and S. Sudarshan

Course Outline and Schedule

Week Mon-Sat	Topics	Text Book	Tests & Assignments
Wk-1 23-Sep	Introduction to Databases Traditional File-Based Systems Database Approach Roles in the Database Environment History of Database Management Systems Advantages and Disadvantages of DBMS Approach	(A) Chap 1	
Wk-2 30-Sep	Database System Concepts and Architecture Database Architecture The Three-Level ANSI-SPARC Architecture: External Level, - The Three-Level Architecture - Schemas, Mappings, and Instances - Data Independence Database Languages Centralized and Client/Server Architectures for DBMSs Functions of a DBMS	(B) Chap 2 (A) Chap 2	
Wk-3 7-Oct	Data Modeling using ER Model High-Level Conceptual Data Models for Database Design Entity Types Relationship Types Attributes Strong and Weak Entity Types Attributes on Relationships Structural Constraints ER Diagrams, Naming Conventions, and Design Issues Relationship Types of Degree Higher than Two	(A) Chap 12 (B) Chap 3	Assignment 1
Wk-4 14-Oct	Relational Data Model and Relational Database Constraints Relational Model Concepts - Relational Data Structure - Database Relations - Properties of Relations - Relational Keys Relational Model Constraints and Relational Database Schemas Update Operations, Transactions, and Dealing with Constraint Violations	(A) Chap 4 (B) Chap 5	Class Test 1
Wk-5 21-Oct	Structured Query Language (SQL) SQL Data Definition and Data Types Specifying Constraints in SQL Basic Retrieval Queries in SQL INSERT, DELETE, and UPDATE Statements in SQL Additional Features of SQL	(A) Chap 6 (B) Chap 6	
Wk-6 28-Oct	Complex SQL Queries, Views and Transactions Aggregate Functions Grouping Views	(A) Chap 6 (B) Chap 7	
Wk-7 4-Nov	Nested Queries Joins Transactions	(A) Chap 7 (B) Chap 7	

Wk-8 11-Nov	The Enhanced ER Model Specialization and Generalization Aggregation Composition	(B) Chap 4 (A) Chap 13	Assignment 2
Wk-9 18-Nov	Mid Term		Mid Term
Wk-10 25-Nov	Relational Algebra Relational Algebra Operations from Set Theory Relational Database Design Using ER-to-Relational Mapping ER-to-Relational Mapping Algorithm	(B) Chap 8,9	Class Test 2
Wk-11 2-Dec	Normalization Functional Dependencies First Normal Form (1NF) Second Normal Form (2NF)	(A) Chap 14 (B) Chap 14	
Wk-12 9-Dec	Third Normal Form (3NF) Boyce–Codd Normal Form (BCNF)	(A) Chap 14,15 (B) Chap 14	
Wk-13 16-Dec	NOSQL Systems Introduction to NOSQL Systems NOSQL Systems and MongoDB	(B) Chap 24	Assignment 3
Wk-14 23-Dec	Transaction Management Transaction Support Concurrency Control Database Recovery	(A) Chap 22	
Wk-15 30-Dec	Big Data and Data Analytics Advanced SQL Stored Procedures Triggers	(C) Chap 10,11	Class Test 3
Wk-16 6-Jan	Revision/ Presentations		
Wk-17 13-Jan	Prep Week		
Wk-18 20-Jan	Final Examination		
29-Jan	Paper Show Day		
31-Jan	Declaration of Result		

Division of Marks:

Class Tests (3-4)	10
Assignments (3-5)	10
Project	15
Mid Term Test	25
Final Exam	40
Total	<u>100</u>

GRADING WILL BE RELATIVE

REVISED GRADING TABLE (For General Reference)			
Range	Percentage	Points	Grade
85.00-100	85.00	4.00	A
81.50-84.99	81.50	3.67	A–
78.00-81.49	78.00	3.33	B+
74.50-77.99	74.50	3.00	B
71.00-74.49	71.00	2.67	B–
67.50-70.99	67.50	2.33	C+
64.00-67.49	64.00	2.00	C
60.50-63.99	60.50	1.67	C–
57.00-60.49	57.00	1.33	D+
50.00-56.99	50.00	1.00	D
0.00-49.99	Below 50	0.00	F